



OPEN Experimental study on pore structure and dynamic mechanical properties of different types of fiber reinforced concrete after elevated temperature action

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Based on the experiment, the effects of elevated temperature (25 °C, 200 °C, 300 °C, 400 °C, 500 °C) and fiber types (SC, CFC, BFC, PPFC, SFC) on the pore structure and dynamic mechanical properties of concrete materials were investigated. The test results show that: (1) Compared with the porosity of specimens at 25 °C, the porosity of plain concrete at 200 °C, 300 °C, 400 °C and 500 °C increases by 14.53%, 26.59%, 45.36% and 81.23%, respectively. Elevated temperature increases the porosity of the specimen, and the addition of fiber can reduce the porosity and porosity complexity of the specimen. (2) At 25 °C, the peak stress of CFC, BFC, PPFC and SFC specimens is 1.44, 1.37, 1.13 and 1.17 times that of plain concrete, and the toughness increase of CFC, BFC, PPFC and SFC specimens is 127.78%, 96.30%, 83.33% and 59.26%, respectively. The addition of fiber can significantly improve the strength and toughness of concrete. (3) Fiber incorporation reduces the energy dissipation effect of concrete specimens, and elevated temperature enhances the energy dissipation effect of concrete specimens. The effect of elevated temperature makes the concrete material deteriorate, and its plastic deformation and damage degree are higher under the impact load. (4) The five kinds of fiber reinforced concrete show the coexistence mode of tensile failure and compressive failure under impact load. The addition of fiber reduces the damage degree of concrete specimens to different degrees, and CF has the best effect on the impact resistance of concrete materials. The research results can provide experimental basis for elevated temperature resistance and dynamic load resistance of different fiber concrete buildings.

Keywords Elevated temperature, Fiber reinforced concrete, Porosity, Fractal dimension, Peak stress, Energy dissipation, Fracture

Fire is one of the disasters with the highest frequency and the most serious loss in nature, and it is a major form of disaster threatening public safety^{1–5}. In common fires, concrete is the building material most directly affected by elevated temperature. In most cases, the structure after fire can be reinforced and repaired, but under the action of elevated temperature, the concrete is prone to burst, resulting in its surface peeling, which affects the structural integrity. Meanwhile, the mechanical properties of concrete materials under the action of elevated temperature will also be affected, thus affecting the safety performance of buildings^{6–10}. Zhu et al.¹¹ carried out experimental studies on the mechanical properties and microstructure of desert sand concrete after elevated temperature, and found that the ultrasonic wave velocity and axial compressive strength of the test pieces decreased after elevated temperature action, the number of internal pores of the test pieces increased, cracks broke through, and the microstructure and macro properties of the test pieces deteriorated due to elevated temperature action. Zhao et al.¹² studied the law of uniaxial tensile and compressive mechanical properties of concrete after elevated temperature, and found that the initiation and expansion time of cracks on the surface of concrete gradually advanced with the increase of temperature, and the expansion degree of cracks became more significant. The tensile strength, compressive strength and elastic modulus of concrete decreased continuously, the strain at the peak stress increased gradually, and the failure characteristics changed from brittleness to ductility. Zhang et

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al.¹³ used 74 mm separated Hopkinson pressure rods to conduct dynamic mechanical properties tests on C30 concrete after cooling at different temperatures, and found that elevated temperature cooling would produce strain softening effect on concrete and reduce the dynamic mechanical properties of specimens, and the crack expansion and fracture degree of specimens increased with the increase of temperature. Hua et al.¹⁴ studied the compressive strength and mesoscopic damage mechanism of concrete after elevated temperature, and found that the compressive strength and elastic modulus of concrete decay slowly at first and then quickly, and the longer the heating time, the higher the attenuation range of strength. The porosity of concrete increased exponentially with the increase of temperature, the number and size of cracks in the specimen increased gradually, and the mechanism of mechanical property damage was caused by dehydration and decomposition of gelling material. Song et al.¹⁵ studied the mechanical properties and fire resistance characteristics of concrete microstructure, and found that the increase of temperature reduced the peak strength of specimens, and the porosity and average pore radius gradually increased with the increase of temperature, and the effect of elevated temperature cracking inside concrete led to the increase of pore volume. The effect of elevated temperature on concrete material will cause damage, thus reducing the safety performance of buildings, so it is very important to take some measures to enhance the high temperature resistance of concrete material in the project.

Existing research results show that adding short fibers to concrete materials can effectively enhance the elevated temperature resistance and mechanical properties of specimens^{16,17}, which provides a new direction for the safety of concrete buildings under elevated temperature. Different types of fiber have different effects on the elevated temperature resistance of concrete materials. In the existing research results, Zhang et al.¹⁸ found that the addition of carbon fiber can enhance the resistance to elevated temperature degradation of light aggregate concrete, and the performance improvement of test pieces is most significant when the volume content of carbon fiber is 0.2%. Sun et al.¹⁹ explored the elevated temperature resistance of glass-polypropylene hybrid fiber concrete through experiments, and found that the fire resistance of the test specimen was significantly improved when the mixture amount of glass fiber and polypropylene fiber was 0.4% and 0.1%, respectively. Zhang et al.²⁰ conducted an experimental study on the elevated temperature resistance of hybrid fiber concrete, and the results showed that the hybrid effect of fiber can significantly improve the elevated temperature resistance of concrete material, making its compressive strength and tensile strength significantly higher than that of plain concrete after elevated temperature. Ge et al.²¹ found through experiments that the addition of PVA fiber and PP fiber can effectively improve the tensile properties of concrete materials after elevated temperature action, and the lifting effect of PVA fiber on concrete materials is higher than that of PP fiber. The effect of different fiber types on the elevated temperature resistance of concrete materials is inconsistent. In construction engineering, carbon fiber (CF), basalt fiber (BF), steel fiber (SF) and polypropylene fiber (PP) are several widely used fibers^{22–25}, and concrete buildings are often subjected to dynamic loads during use^{26–31}. Therefore, it is of great significance to study the microstructure and dynamic mechanical properties of fiber reinforced concrete under elevated temperature.

The different types of fibers have inconsistent degrees of influence on the high-temperature resistance of concrete materials. To further explore the influence laws of high-temperature effects and fiber types on the properties of concrete materials, based on the current research results at home and abroad, this paper conducts experimental studies on the microscopic pore structure and dynamic mechanical properties of CF concrete, BF concrete, SF concrete and PPF concrete after different high-temperature effects. The effects of fiber type and temperature on T_2 map, porosity and evolution of mechanical properties of concrete under impact load were analyzed. The test process is shown in Fig. 1. The research results can fill the gap in the field of study on the performance evolution law of fiber concrete under dynamic loading after high-temperature action.

Test Materials

Conch P·O 42.5 ordinary Portland cement is selected as the raw material for concrete preparation, and some parameters are shown in Table 1. The fine aggregate is made of natural river sand with fineness modulus of 2.65 from Huaihe River and its own density is 2650 kg·m⁻³. The coarse aggregate is made of granite with a particle size of 5–15 mm. Water selection laboratory tap water, admixture polycarboxylic acid superplasticizer, water reduction efficiency of 25%. The mix ratio of each material of concrete is cement: Water: fine aggregate: coarse aggregate: water reducing agent = 1:1.25:2.3:0.4. The fibers used were CF, BF, PP and SF, in which CF length was 10 mm and the volume content was 0.24%³². BF length is 18 mm, volume content is 0.1%³³, PP length is 18 mm, mass content is 0.6 kg/m³³⁴. The SF length is 13 mm, the volume content is 1.5%³⁵, and the four fibers are shown in Fig. 2. Some of the performance is shown in Table 2. The plain concrete and four kinds of fiber concrete were poured according to the matching ratio, and the curing treatment was carried out for 28 days in the standard curing room (temperature of 20 ± 2 °C, relative humidity ≥ 95%) after the pouring was completed.

Test scheme and device

The temperature of low temperature fire is below 200 °C, the temperature of medium temperature fire is between 200 °C–500 °C, and the temperature of elevated temperature fire is above 500 °C, so the temperature of this test is set to 25 °C and 500 °C. After reaching the curing age, the plain concrete and four kinds of fiber concrete specimens were treated with elevated temperature, and the numbered specimens were placed in the SX-12-5.0 Muffle furnace for elevated temperature heating. The rated power of the instrument was 6 kW, the maximum temperature was 1200 °C, and the heating rate was 5 °C/min. After rising to the predetermined temperature, constant temperature treatment was carried out for 6 h, and the temperature was set at 25 °C, 200 °C, 300 °C, 400 °C and 500 °C, respectively. After the elevated temperature, the specimen was naturally cooled to room temperature³⁶. The micro-pore structure and dynamic mechanical properties of all high-temperature specimens were studied after processing.

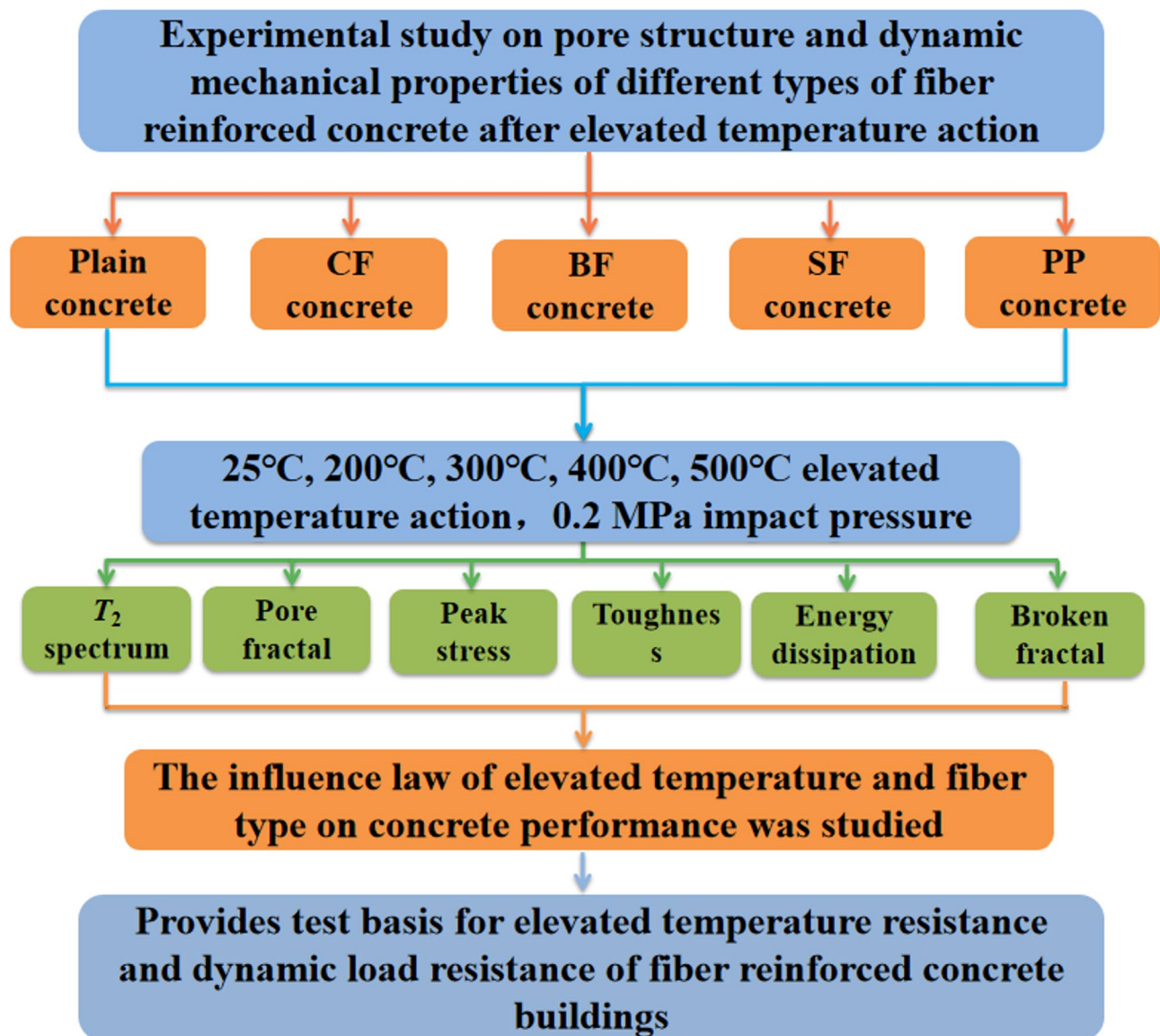


Fig. 1. Test process.

Density (kg/cm ³)	BET (m ² /g)	The proportion of chemical composition (%) (XRF)					
		CaO	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	SO ₃	MgO
1.91	1.48	31.31	1.94	0.9	0.23	43.49	0.29

Table 1. Portland cement parameter.

AVANCE NEO NMR device was used to detect the pore structure of fiber reinforced concrete samples after elevated temperature action, in order to obtain the effect of fiber types and elevated temperature action on the pore structure of concrete materials. When using low magnetic field to analyze the internal pore structure of materials, the relaxation time mainly depends on the surface relaxation, the surface relaxation expression is as follows^{37,38}:

$$\frac{1}{T_2} = \rho \left(\frac{S}{V} \right)_{\text{pore}} \quad (1)$$

where: T_2 is the transverse relaxation time, ms; ρ is the transverse surface relaxation strength, $\mu\text{m}/\text{ms}$, and $\frac{S}{V}$ is the specific pore area, μm^{-1} .

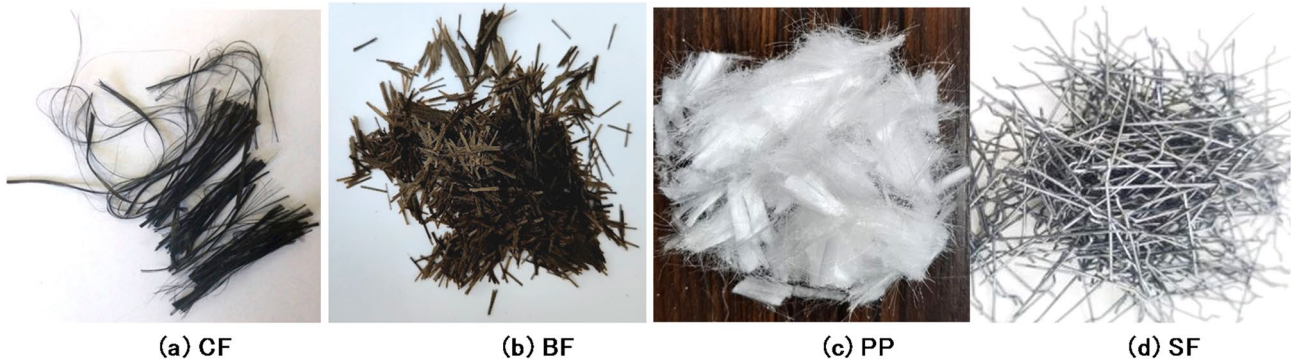


Fig. 2. Four fibers.

Fibre type	Diameter (μm)	Length (mm)	Modulus of elasticity (GPa)	Density (g cm ⁻³)	Tensile strength (MPa)
CF	8	10	180	1.75	3000
BF	18	18	102	2.65	3800
PPF	30	18	15	0.90	550
SF	600	13	210	7.80	1500

Table 2. Performance parameters of carbon fiber.

When the internal pores of the specimen can be simplified into a spherical or columnar pipe pore structure, T_2 in Eq. (1) is the transverse relaxation time, which can be further converted into the relationship between the pore radius:

$$\frac{1}{T_2} = \frac{\rho}{r} F_s \tag{2}$$

where: r is the pore radius, μm ; F_s is a geometric form factor (spherical $F_s=3$, cylindrical $F_s=2$).

The fractal dimension can be used to quantitatively characterize the complexity of irregular and disordered pore shapes in concrete. The larger the fractal dimension, the stronger the heterogeneity of its internal pore distribution. According to the geometry principle, the internal pore distribution of concrete materials conforms to the fractal structure³⁹. The relationship between the number of pores N and r whose pore diameter is larger than r is satisfied:

$$N(\geq r) = \int_r^{r_{\max}} P(r)dr = ar^{-D} \tag{3}$$

where, $r_{\max}$ is the maximum pore radius, $P(r)$ is the density function of pore distribution, a is the scale function, and D is the fractal dimension of pore structure.

In the nuclear magnetic resonance test $r = \rho_2 F_s T_2$, T_2 is the relaxation time, which can be obtained by transformation:

$$S_V = (T_{2\max}/T_2)^{D-3} \tag{4}$$

Taking the logarithm of both sides of the equation:

$$\lg S_v = (3 - D) \lg T_2 + (D - 3) \lg T_{2\max} \tag{5}$$

Impact compression test was carried out on the specimen with a separate Split Hopkinson Pressure Bar (SHPB) device with a diameter of 74 mm. The test device is shown in Fig. 3. After the processing of all high-temperature specimens, the dynamic impact loading pre-test was carried out on the specimens, and the impact pressure was finally selected as 0.2 MPa. The stress, strain, strain rate and energy of the specimen were calculated by three-wave method. The obtained waveform and typical stress balance curve are shown in Fig. 4, and the calculation formula is as follows^{40,41}:



Fig. 3. SHPB installation.

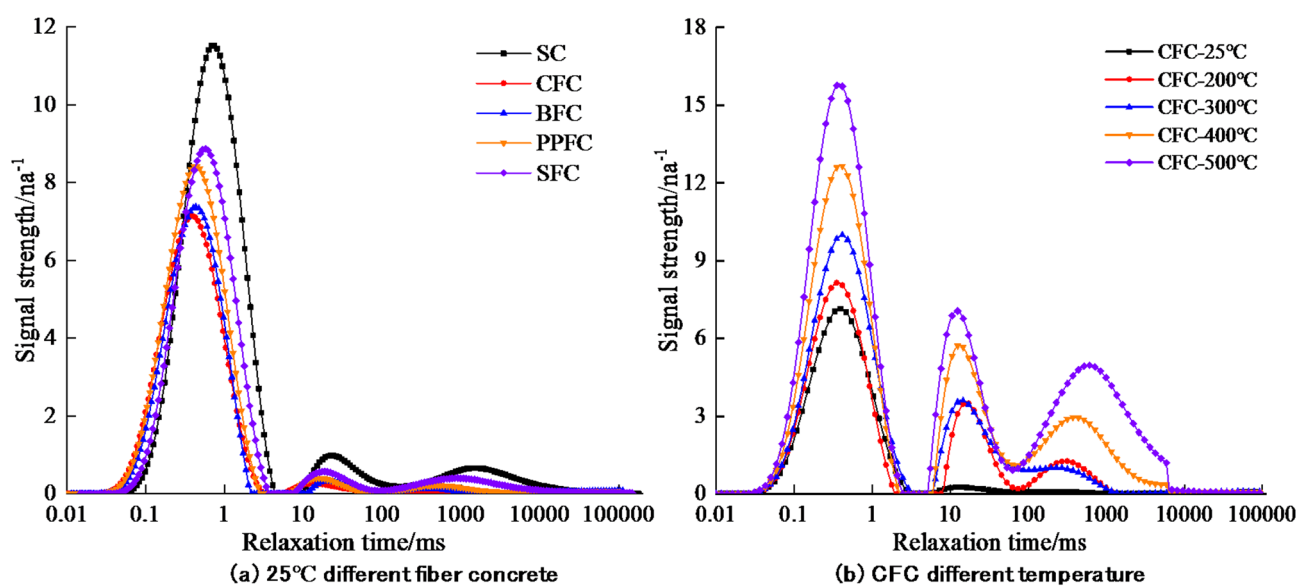


Fig. 4. T_2 map distribution.

$$\begin{cases} \dot{\varepsilon}(t) = \frac{C_0}{l} [\varepsilon_i(t) - \varepsilon_r(t) - \varepsilon_t(t)] \\ \varepsilon(t) = \frac{C_0}{l} \int_0^t [\varepsilon_i(t) - \varepsilon_r(t) - \varepsilon_t(t)] dt \\ \sigma(t) = \frac{AE}{2A_s} [\varepsilon_i(t) + \varepsilon_r(t) + \varepsilon_t(t)] \end{cases} \quad (6)$$

$$\begin{cases} W_i(t) = AEC_0 \int_0^t \varepsilon_i^2(t) dt \\ W_r(t) = AEC_0 \int_0^t \varepsilon_r^2(t) dt \\ W_t(t) = AEC_0 \int_0^t \varepsilon_t^2(t) dt \end{cases} \quad (7)$$

$$W_s(t) = W_i(t) - W_r(t) - W_t(t) \quad (8)$$

$$\varepsilon_d = \frac{W_s(t)}{V} \quad (9)$$

where $\dot{\varepsilon}(t)$, $\varepsilon(t)$, and $\sigma(t)$ are the strain rate, strain and stress, respectively. $\varepsilon_i(t)$, $\varepsilon_r(t)$, and $\varepsilon_t(t)$ are the incident strain, reflected strain, and transmitted strain, respectively. l is the height. C_0 is the longitudinal wave velocity. E is the modulus of elasticity. A is the cross-sectional area, and A_s is the cross-sectional area of the specimen. W_i , W_r , W_t , and W_s are the incident, reflected, transmitted, and dissipated energy. ε_d is the crushing energy dissipation density, and V is the volume.

Results

T_2 spectrum

The T_2 spectrum curve value detected by NMR represents the NMR signal represented by pore water inside the specimen. The T_2 spectrum distribution of five different concrete specimens at 25°C is shown in Fig. 4(a). The T_2 map distribution of carbon fiber concrete under different elevated temperature effects is shown in Fig. 4(b), and the main peak area and proportion of T_2 map are shown in Table 3.

Table 1; Fig. 4 show that the T_2 map curves of different types of fiber concrete after elevated temperature action all show a trimodal shape, and the peak value of the main peak is much higher than the secondary peak and trimodal peak value. Figure 4(a) shows that the addition of fiber can significantly reduce the peak value of the spectrum. Compared with the peak value of plain concrete at 25°C, the peak value of the main peak of CFC, BFC, PPFC and SFC specimens decreases by 38.11%, 35.85%, 26.39% and 23.00% respectively. The addition of fiber can improve the internal pore structure of concrete during pouring and curing, fill the internal primary pores, and thus reduce the porosity of specimens. In addition, the improvement of pore structure of specimens by CFC is higher than that by BFC and higher than that by PPFC, indicating that CF has the best effect on improving the internal structure of concrete, while SF has the worst effect on improving the internal pore structure of specimens. The addition of the same fiber can reduce the peak of the two peaks and three peaks, reduce the relaxation time, and make the pore diameter and the number of pores in the concrete. Figure 4(b) shows that the peak values of T_2 spectra of CFC specimens are increased due to elevated temperature. Compared with 25°C, 200°C, 300°C, 400°C and 500°C, the increases of the main peak value were 14.31%, 39.83%, 77.98% and 121.46%, and the secondary peak value was 12.35, 13.92, 22.12 and 27.31 times that of the peak value at 25°C. The results show that the porosity and the number of pores in the concrete are greatly increased by the elevated temperature, and the higher the temperature, the more significant the increase in the porosity. On the one hand, as the temperature increases, the water inside the specimen evaporates, the bonding between aggregates decreases, the number of cracks increases, and the specimen itself is damaged. On the other hand, the volume expansion and contraction of aggregates during the high-temperature cooling process causes the internal cracks to regrow and expand, thus further increasing the internal porosity of the specimen. The higher the temperature, the stronger this process is, and the more significant the pore growth is.

Porosity and pore fractal dimension

The porosity changes of five different concrete specimens after different elevated temperatures are shown in Fig. 5, and the calculation method of fractal dimension is shown in Fig. 6. The fractal dimension changes of specimens under different elevated temperature effects are shown in Fig. 7.

Figure 5 shows that the porosity of fiber concrete specimens is always lower than that of plain concrete under different elevated temperatures, and the addition of several fibers can significantly reduce the porosity of specimens. Among them, the improvement effect of carbon fiber and basalt fiber on the internal pore structure of concrete specimens is higher than that of polypropylene fiber and steel fiber. With the increase of temperature, the porosity of the samples increased continuously. Compared with the porosity of the samples at 25°C, the porosity of the concrete at 200°C, 300°C, 400°C and 500°C increased by 14.53%, 26.59%, 45.36% and 81.23%, respectively. The increase of porosity of carbon fiber concrete was 35.46%, 52.48%, 94.33% and 141.84%, and that of basalt fiber concrete was 19.27%, 32.98%, 57.17% and 90.36%, respectively. The porosity of polypropylene

Specimen number	Principal peak			Secondary peak		
	Peak value (na ⁻¹)	Relative area	Proportion (%)	Peak value (na ⁻¹)	Relative area	Proportion (%)
SC-25	11.52	71.99	89.56	0.97	4.64	5.77
SC-200	12.89	92.44	54.87	4.78	54.67	32.45
SC-300	15.44	102.49	53.22	6.22	66.59	34.58
SC-400	17.02	111.95	50.78	8.87	74.71	33.89
SC-500	21.56	130.62	48.58	10.12	88.38	32.87
CFC-25	7.13	44.89	93.22	0.26	2.40	4.98
CFC-200	8.15	54.79	63.28	3.47	23.84	27.54
CFC-300	9.97	66.05	68.89	3.62	22.60	23.57
CFC-400	12.69	77.63	64.52	5.75	31.98	26.58
CFC-500	15.79	99.42	66.78	7.10	30.44	20.45
BFC-25	7.39	45.33	92.26	0.31	2.14	4.35
BFC-200	8.45	62.79	65.78	3.57	24.40	25.56
BFC-300	10.00	69.67	63.22	4.21	26.96	24.46
BFC-400	12.54	92.09	67.88	6.87	40.13	29.58
BFC-500	15.54	97.53	60.87	7.54	48.95	30.55
PPFC-25	8.48	52.41	93.77	0.38	2.27	4.06
PPFC-200	9.87	66.02	59.88	3.98	31.71	28.76
PPFC-300	13.44	85.02	57.65	5.84	39.63	26.87
PPFC-400	16.01	111.46	58.98	7.65	56.49	29.89
PPFC-500	19.54	124.58	53.65	9.57	70.45	30.34
SFC-25	8.87	55.49	91.01	0.58	3.18	5.22
SFC-200	9.97	72.17	65.47	3.78	22.67	20.56
SFC-300	12.78	94.19	63.28	5.55	28.10	18.88
SFC-400	15.23	91.15	59.78	6.84	39.17	25.69
SFC-500	18.12	107.78	60.22	8.12	48.92	27.33

Table 3. The area and proportion of main peak in T_2 atlas. SC-25: SC stands for plain concrete, 25 stands for temperature of 25 °C. CFC, BFC, PPFC and SFC stand for carbon fiber concrete, basalt fiber concrete, polypropylene fiber concrete and steel fiber concrete respectively.

fiber concrete increased by 20.00%, 42.22%, 67.18% and 120.00%, while that of steel fiber concrete increased by 13.58%, 28.43%, 44.57% and 75.40%, respectively. As the temperature rises, the water inside the specimen evaporates, cracks expand and pores are formed, which leads to the increase of porosity. The increase of porosity of CFRP and PP fiber concrete is higher than that of basalt fiber concrete and steel fiber concrete. This is because the melting point of carbon fiber and polypropylene fiber is lower than the other two fibers, which makes it easier to deform and destroy the fiber under elevated temperature, and the increase in porosity will be more significant.

Figure 6 shows the calculation method to characterize the fractal characteristics of the internal pores of the specimen. The fractal dimension D_{min} is positively correlated with the specific surface area and volume of the concrete micropores. The fractal dimension D_{max} is positively correlated with the specific surface area and pore volume of medium and large pores. The fractal dimension of pores reflects the irregularity and complexity of the internal pore structure of concrete. The smaller the pore fractal dimension is, the less irregular the pore structure is and the more simple the pore structure is. The larger the value, the more irregular the pore structure and the more complex the pore structure^{42,43}. At the same time, the fractal dimension of pores also indicates the structural characteristics of pores. The more pores and micropores in the pore medium, the greater the degree of irregularity and irregularity of the pore surface, resulting in the increase of the fractal dimension of the pore structure. Figure 7 shows that D_{min} and D_{max} of the five concrete specimens increase continuously with the increase of temperature, and D_{max} of the specimens is always higher than D_{min} . It shows that the large pore structure inside the specimen is more complex. Compared with the fractal dimension of internal pores at 25 °C, D_{min} increases by 0.015, 0.04, 0.089 and 0.101, respectively, and D_{max} increases by 0.052, 0.035, 0.021 and 0.024 at 200 °C, 300 °C, 400 °C and 500 °C, respectively. At 25 °C, D_{min} of plain concrete was higher than that of CFC, BFC, PPFC and SFC specimens by 0.101, 0.097, 0.055 and 0.066, respectively, and at 500 °C, D_{min} of plain concrete was higher than that of CFC, BFC, PPFC and SFC specimens by 0.076, 0.064, 0.027 and 0.043, respectively. With the increase of temperature, the number and diameter of the pores inside the specimen increase, and the complexity of the structure increases, while the effect of elevated temperature will reduce the gap of the complexity of the pore structure between the plain concrete and the fiber concrete.

Peak stress and toughness

The stress-strain curves of 5 kinds of concrete specimens at 25 °C are shown in Fig. 8(a), and the relationship between peak stress and temperature is shown in Fig. 8(b). Toughness can be used to quantitatively characterize

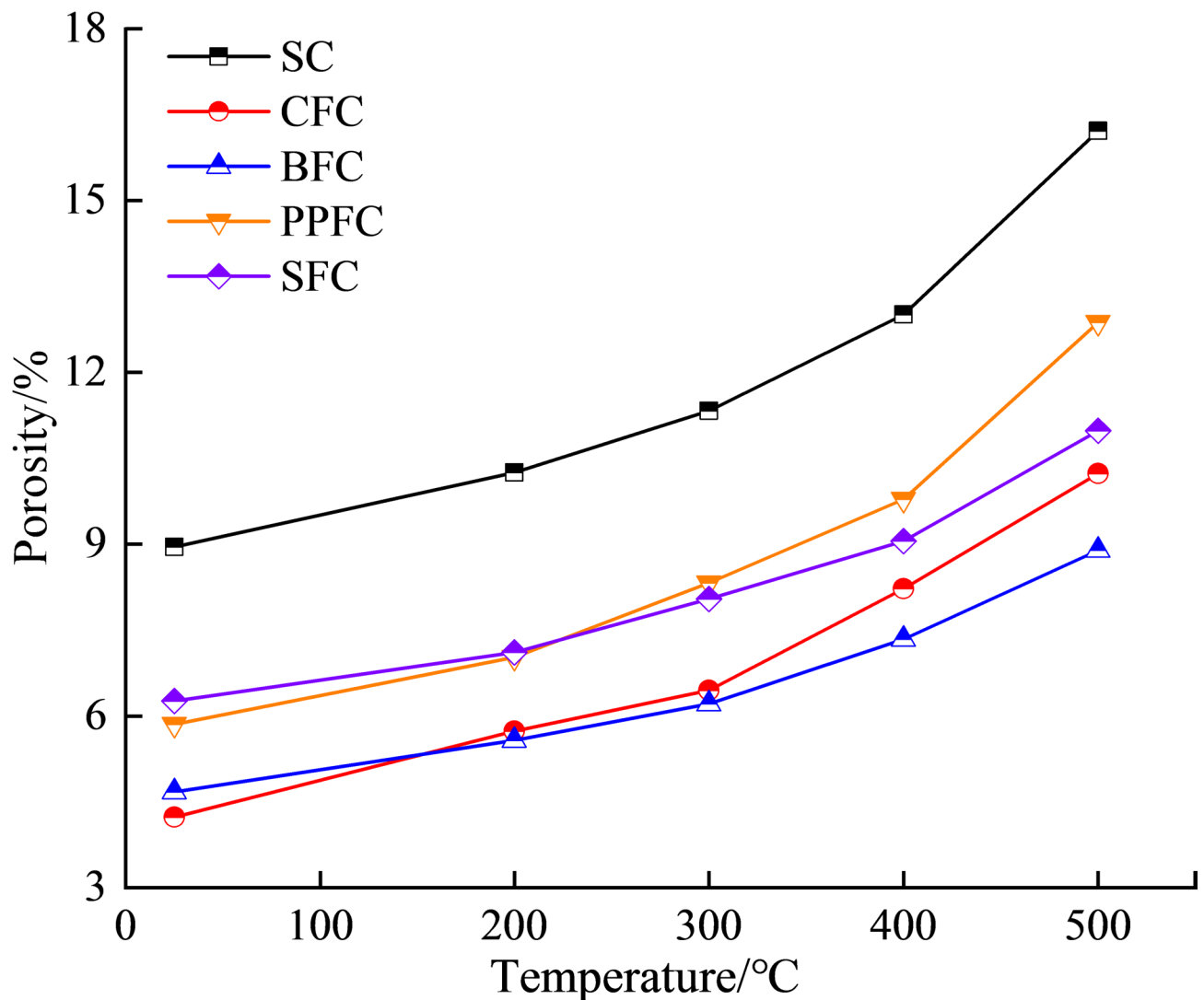


Fig. 5. Variation of porosity with temperature.

the deformation ability of materials under load. The change of toughness of 5 concrete specimens under elevated temperature is shown in Fig. 9.

Figure 8(a) shows that the peak stress and deformation degree of fiber reinforced concrete are significantly higher than that of plain concrete, and the peak stress of CFC, BFC, PPFC and SFC at 25°C is 1.44, 1.37, 1.13 and 1.17 times that of plain concrete. The addition of fiber can effectively improve the compressive strength of concrete specimens. On the one hand, due to the high tensile strength of the fiber itself, it can withstand part of the external load when subjected to external load, thus enhancing the compressive performance of the specimen as a whole. On the other hand, the fiber bar forms a spatial skeleton structure inside the concrete for support, and there is good cohesion between the fiber and the aggregate, which effectively improves the integrity of the concrete material, thus enhancing the compressive strength and plastic deformation ability of the specimen. The effect of several fibers on the strength of concrete is inconsistent, among which the effect of carbon fiber on the strength of concrete is the largest, and the effect of polypropylene fiber on the strength of concrete is the least. Figure 8(b) shows that the peak stress of several fiber reinforced concrete will decrease significantly after elevated temperature action. Compared with the peak stress at 25°C, the peak stress of SC specimen at 200°C, 300°C, 400°C and 500°C decreased by 5.03%, 12.40%, 25.30% and 44.04%, respectively. The peak stress reduction of CFC specimens was 3.26%, 10.72%, 1.07% and 24.36%, and that of BFC specimens was 3.49%, 9.75%, 17.75% and 27.04%, respectively. The peak stress reduction of PPFC specimens was 4.87%, 13.28%, 24.19% and 39.79%, and that of SFC specimens was 1.98%, 8.19%, 16.39% and 24.93%, respectively. When the temperature increase is small, the water vapor inside the specimen forms and escapes, and the evaporation and escape of water causes dehydration reaction of hydrated calcium silicate, resulting in the new formation of pores and the expansion of cracks, and the mechanical properties show a decrease in peak strength. With the further increase of temperature, the strength of the specimen decreased significantly, which was because the hydration product $\text{Ca}(\text{OH})_2$ decomposed with the further increase of temperature, and the C-S-H gelling decomposed

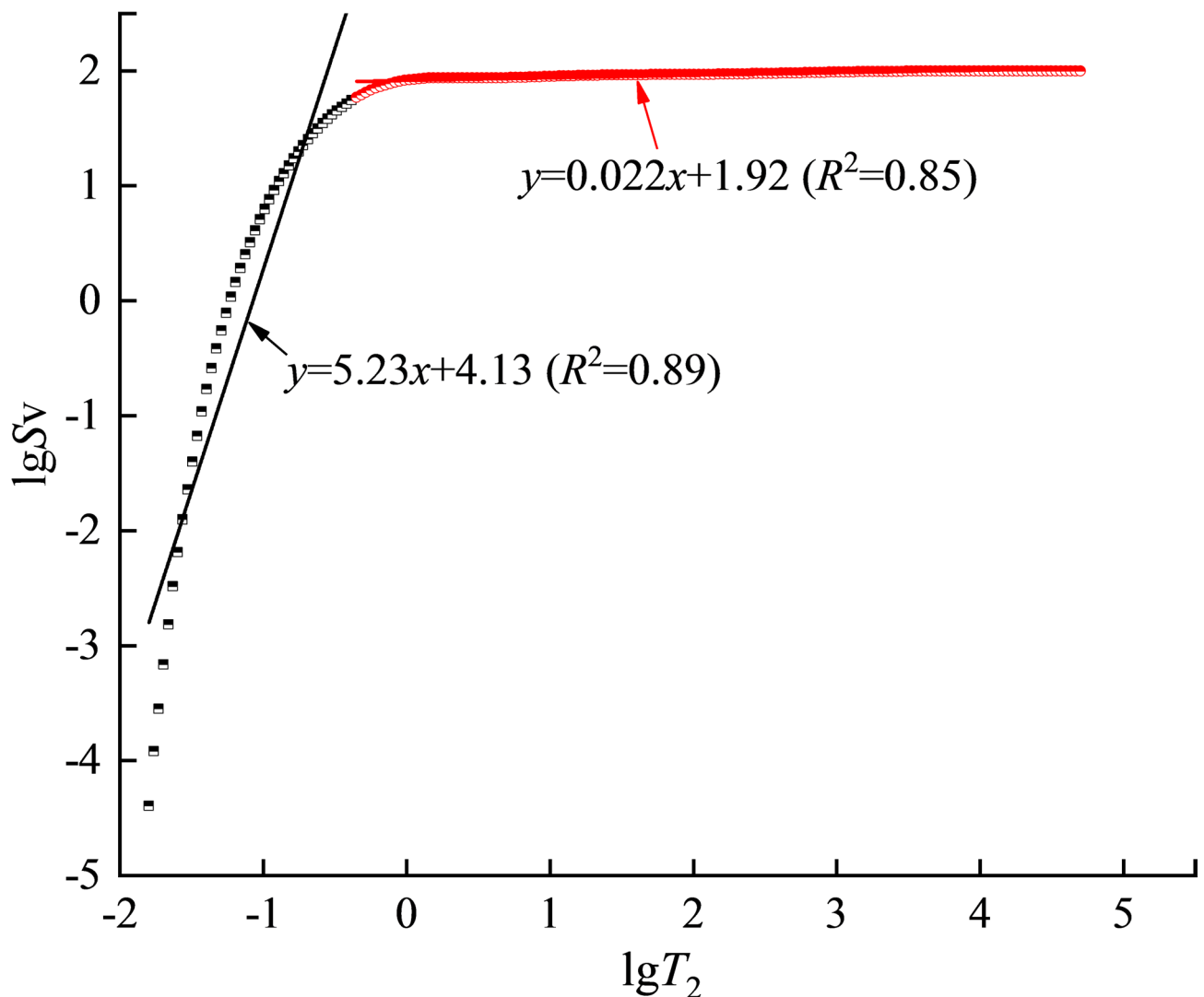


Fig. 6. The calculation method of fractal dimension.

after water loss. The reduction of gelling material reduced the cohesion between aggregates. At the same time, the free water generated by decomposition continues to transform into water vapor, the internal vapor pressure increases, the volume shrinkage of cement slurry accompanied by the decomposition of hydration products and the thermal expansion of aggregate under elevated temperature conditions accelerate the expansion of cracks to a certain extent, so that the macroscopic micro-cracks appear on the concrete surface. When the temperature reaches more than 500 °C, it is due to the severe decomposition of C-S-H gel, the hydration product in the concrete cement slurry, which almost completely loses the bound water and transforms into a large number of C₂S without cementing ability, and obvious expansion cracks occur on the macroscopic surface of the specimen, and the strength of the concrete specimen decreases accordingly. Comparing the peak stress reduction of several specimens, it can be seen that the stress reduction of CFC specimens under elevated temperature is the smallest, followed by SFC, BFC and PPFC, and the stress reduction of SC specimens is the largest, indicating that the addition of fiber can enhance the elevated temperature resistance of concrete materials, and CF has the best effect on the elevated temperature resistance of concrete materials. The addition of fiber can enhance the bonding between aggregates, resist the deformation degree under elevated temperature, and slow down the stress reduction of the specimen under elevated temperature. CF, BF, SF fibers have a high melting point, CF melting point up to 800 °C, BF melting point up to 1500 °C, SF melting point up to 1500 °C, the performance can maintain good under elevated temperature. Therefore, these three kinds of fiber concrete specimens have good elevated temperature resistance, low melting point of PPF, and easy deformation and degradation under elevated temperature action. Therefore, their effect on the improvement of elevated temperature resistance of concrete materials is limited.

The toughness of concrete represents its deformation resistance. Figure 9 shows that the addition of several fibers can improve the toughness of concrete materials to different degrees. Compared with the toughness of SC specimens at 25 °C, the toughness of CFC, BFC, PPFC and SFC specimens increased by 127.78%, 96.30%, 83.33% and 59.26%, respectively. Combined with the stress-strain curve of the specimen in Fig. 8(a), it can be

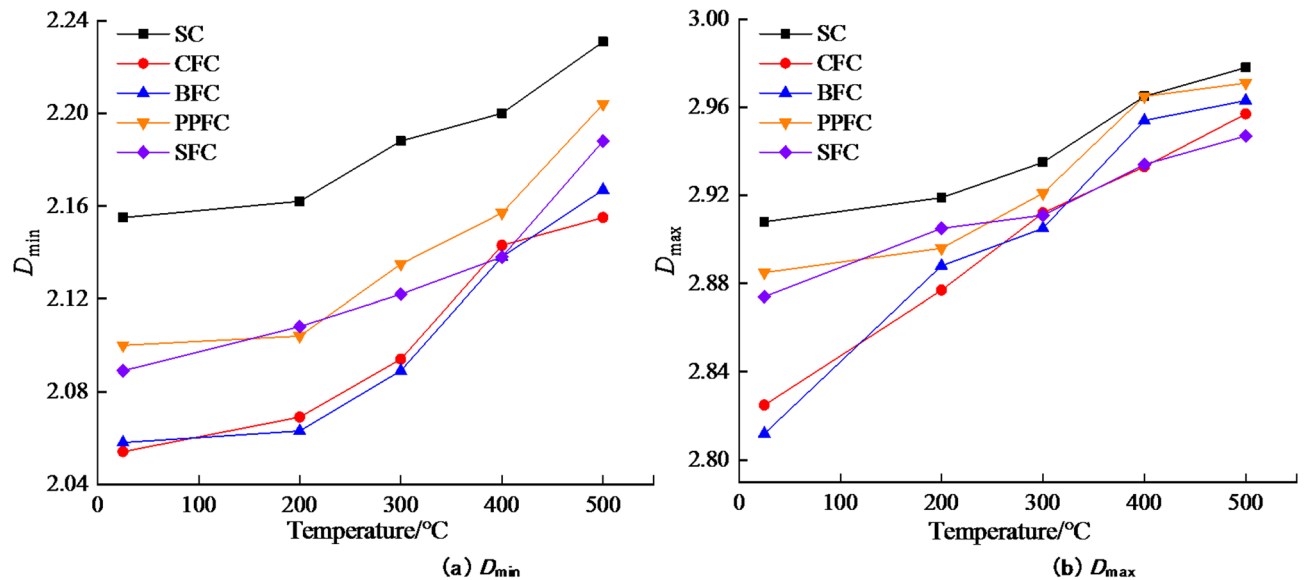


Fig. 7. The relationship between fractal dimension and temperature.

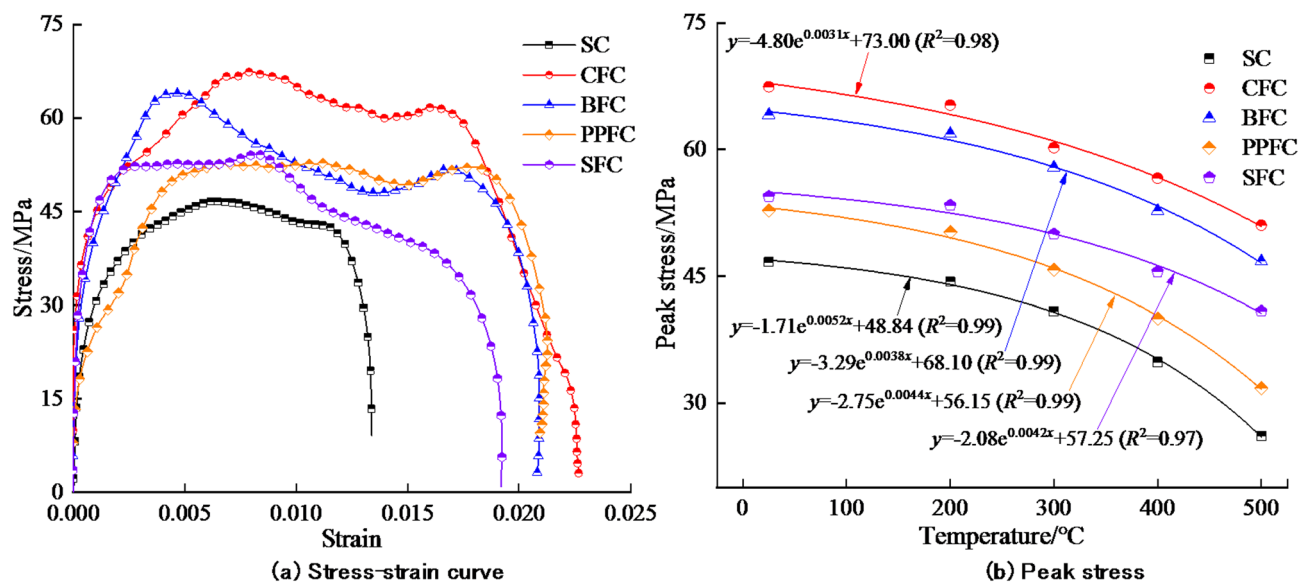


Fig. 8. Effect of fiber type and temperature on specimen stress.

seen that the addition of fiber not only increases the peak stress of the specimen, but also significantly increases its deformation. The addition of fiber not only enhances the impact load resistance of concrete material, but also significantly improves the ductility of the material under load, making concrete buildings change from brittle failure to ductile failure when subjected to dynamic load, which will greatly improve the safety performance of buildings. Among the four fibers, CF improves the toughness of concrete materials the most at 25°C, indicating that carbon fiber has the best effect on the dynamic load resistance of concrete, followed by BF, PPF and SF. The toughness of the five concrete specimens decreased linearly with the increase of temperature. Compared with the toughness of the specimen at 25°C, the toughness of the SC specimen at 200°C, 300°C, 400°C and 500°C decreased by 5.56%, 16.67%, 29.63% and 44.44%, respectively. The peak stress reduction of CFC specimens was 3.25%, 12.20%, 22.76% and 34.96%, and that of BFC specimens was 7.55%, 15.09%, 26.42% and 38.68%, respectively. The peak stress decreases of PPFC specimens were 10.10%, 24.24%, 36.36%, 52.53%, and that of SFC specimens were 2.33%, 9.30%, 18.60%, 27.91%, respectively. The toughness of PPFC specimen decreased the most, and that of SFC specimen decreased the least. The elevated temperature resistance of PPF is poor, and the deterioration degree is high under the action of elevated temperature, so the mechanical properties of PPFC drop the most. SF has good elevated temperature resistance, and elevated temperature has little influence on its performance, so the mechanical properties of SFC specimens are less affected by elevated temperature.

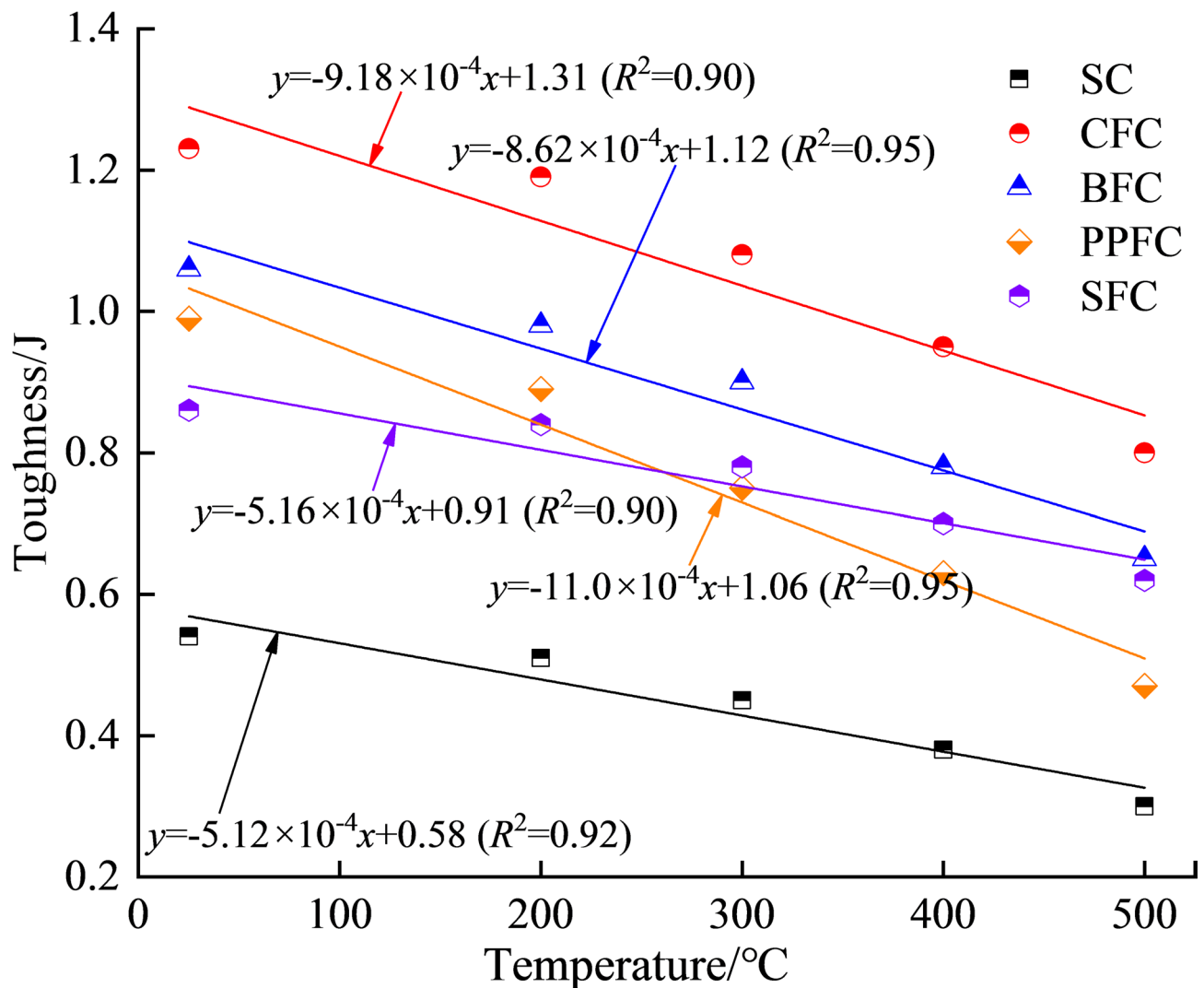


Fig. 9. Variation of toughness with temperature.

Energy dissipation

The energy changes of fiber reinforced concrete specimens under different elevated temperatures are shown in Table 4. The relationship between energy and temperature of each specimen is shown in Fig. 10, and the fitting relationship between crushing dissipated energy density and temperature is shown in Fig. 11.

According to Table 2; Fig. 10, when the external input energy is the same under impact load, the reflected energy and dissipative energy of plain concrete are higher than that of fiber concrete, while the transmitted energy is lower than that of fiber concrete. Based on the one-dimensional stress wave propagation theory and fiber reinforcement mechanism, it can be seen that the stress wave will be reflected at the contact surface of the bar to the specimen due to the different wave impedance of the material. Part of the external input energy is reflected back to the incident rod, part is transmitted to the transmission rod, and the rest is consumed by the deformation and failure of the concrete material. When the wave impedance difference between the bar and the specimen is larger, the reflection energy is larger and the transmission energy is smaller. The inclusion of fiber in concrete will enhance the integrity of the specimen and reduce the internal porosity, which will reduce the wave impedance difference between the concrete and the bar, increase the transmission energy and reduce the reflection energy. The dissipative energy is used for plastic deformation and fracture of concrete materials, and the inclusion of fiber reduces the dissipative energy of specimens. It is also due to the fiber incorporation that the impact resistance of the specimen is enhanced, and the plastic deformation and damage degree of the specimen are reduced after loading, and the dissipated energy is also reduced. As the temperature increases, the reflection energy and dissipative energy of the specimen increase, while the transmission energy decreases. Elevated temperature will cause damage to the concrete material, reduce the wave impedance and impact deformation resistance of the specimen, reduce the ability to penetrate, and increase the plastic deformation of the specimen.

In order to further quantitatively characterize the energy dissipation effect of concrete materials, the energy dissipation density per unit volume of specimens was calculated. Figure 11 shows that the inclusion of fiber reduces the crushing energy dissipation density of concrete specimens. Compared with the crushing energy

Specimen number	Mean strain rate (s ⁻¹)	Incident energy (J)	Reflected energy (J)	Transmitted energy (J)	Dissipated energy (J)	Crushing energy density (J cm ⁻³)
SC-25	85.32	127.8	50.45	41.77	35.58	0.224
SC-200	87.38	130.5	53.32	39.36	37.82	0.238
SC-300	91.33	128.6	56.45	29.48	42.67	0.268
SC-400	94.88	125.9	57.67	22.45	45.78	0.288
SC-500	97.21	127.7	60.87	17.62	49.21	0.309
CFC-25	65.6	129.2	40.78	61.97	26.45	0.166
CFC-200	67.9	131.2	42.52	60.65	28.03	0.176
CFC-300	70.7	126.4	46.62	46.93	32.85	0.207
CFC-400	73.8	127.7	48.03	43.13	36.54	0.230
CFC-500	77.5	129.8	49.89	39.66	40.25	0.253
BFC-25	67.9	128.6	41.85	58.98	27.77	0.175
BFC-200	70.8	129.1	42.58	57.20	29.32	0.184
BFC-300	73.8	127.7	45.85	46.97	34.88	0.219
BFC-400	77.8	128.3	48.87	39.78	39.65	0.249
BFC-500	80.2	126.9	50.48	33.10	43.32	0.272
PPFC-25	71.9	127.4	44.58	53.80	29.02	0.182
PPFC-200	73.1	128.3	48.89	47.09	32.32	0.203
PPFC-300	76.7	132.2	52.65	41.57	37.98	0.239
PPFC-400	81.7	131.5	57.77	31.22	42.51	0.267
PPFC-500	83.8	130.2	60.01	22.34	47.85	0.301
SFC-25	74.4	129.69	44.78	55.03	29.88	0.188
SFC-200	75.6	131.22	47.21	51.80	32.21	0.203
SFC-300	79.9	128.88	50.65	41.49	36.74	0.231
SFC-400	83.9	130.02	53.38	36.43	40.21	0.253
SFC-500	85.3	131.54	56.65	29.04	45.85	0.288

Table 4. The energy changes of fiber reinforced concrete specimens under different elevated temperatures.

density of SC specimen at 25 °C, the crushing energy density of CFC, BFC, PPFC and SFC specimen decreased by 25.89%, 21.88%, 18.75% and 16.07%, respectively. At 500 °C, the energy dissipation density of CFC, BFC, PPFC and SFC specimens decreased by 18.12%, 11.97%, 2.59% and 6.80%, respectively. The addition of fiber enhances the anti-deformation ability of concrete material, and the deformation and damage degree of fiber concrete decreases under impact load, and the energy used for specimen deformation, failure and crack propagation decreases accordingly. With the increase of temperature, the energy dissipation density of the five concrete specimens increases as a quadratic function. Compared with the energy dissipation density of the specimen at 25 °C, the energy dissipation density of SC at 200 °C, 300 °C, 400 °C and 500 °C increased by 6.25%, 19.64%, 28.57% and 37.95%, respectively. The energy consumption density of CFC at 200 °C, 300 °C, 400 °C and 500 °C increased by 6.02%, 24.70%, 38.55% and 52.41%, respectively. At 200 °C, 300 °C, 400 °C and 500 °C, the increase of BFC crushing energy density was 5.14%, 25.14%, 42.29% and 55.43%, respectively. PPFC crushing energy density increases by 11.54%, 31.32%, 46.70% and 65.38% at 200 °C, 300 °C, 400 °C and 500 °C, respectively. The energy consumption density of SFC at 200 °C, 300 °C, 400 °C and 500 °C increased by 7.98%, 22.87%, 34.57% and 53.19%, respectively. The effect of elevated temperature makes the concrete structure deteriorate, and the specimen produces a greater degree of plastic deformation and failure under the external impact load, and more external input energy is consumed by the crack expansion and failure of the specimen, and the dissipative energy increases accordingly. Among them, the dissipation energy of PPFC increases most significantly, which is because PP is easy to deform or even melt under the action of high temperature, losing the role of fiber in the concrete interior, and the dissipative energy of the specimen increases more significantly.

Fracture fractal characteristics

The fracture and crushing morphology of five kinds of concrete under impact load at 25 °C is shown in Fig. 12, and the fracture and crushing morphology of plain concrete and carbon fiber concrete under different high temperature action is shown in Fig. 13.

Figure 12 shows that under the impact load, the five kinds of fiber concrete present a coexistence mode of tensile failure and compressive failure, and the addition of fiber reduces the failure degree of concrete specimens to different degrees. Among them, the CFC specimens have the largest broken block, the most broken number, and the lowest damage degree, which indicates that CF has the best effect on the impact resistance of concrete materials. The internal porosity of plain concrete is high, and the content of fiber breaks down part of the pore structure to make the cement slurry through and full, thus reducing the total porosity of the specimen and strengthening the integrity of the specimen. There is good compatibility between fiber and cement-based material, and there is good bonding property between them. The fibers are randomly distributed inside the concrete to form a spatial network structure, thereby limiting the expansion cracking of concrete and improving

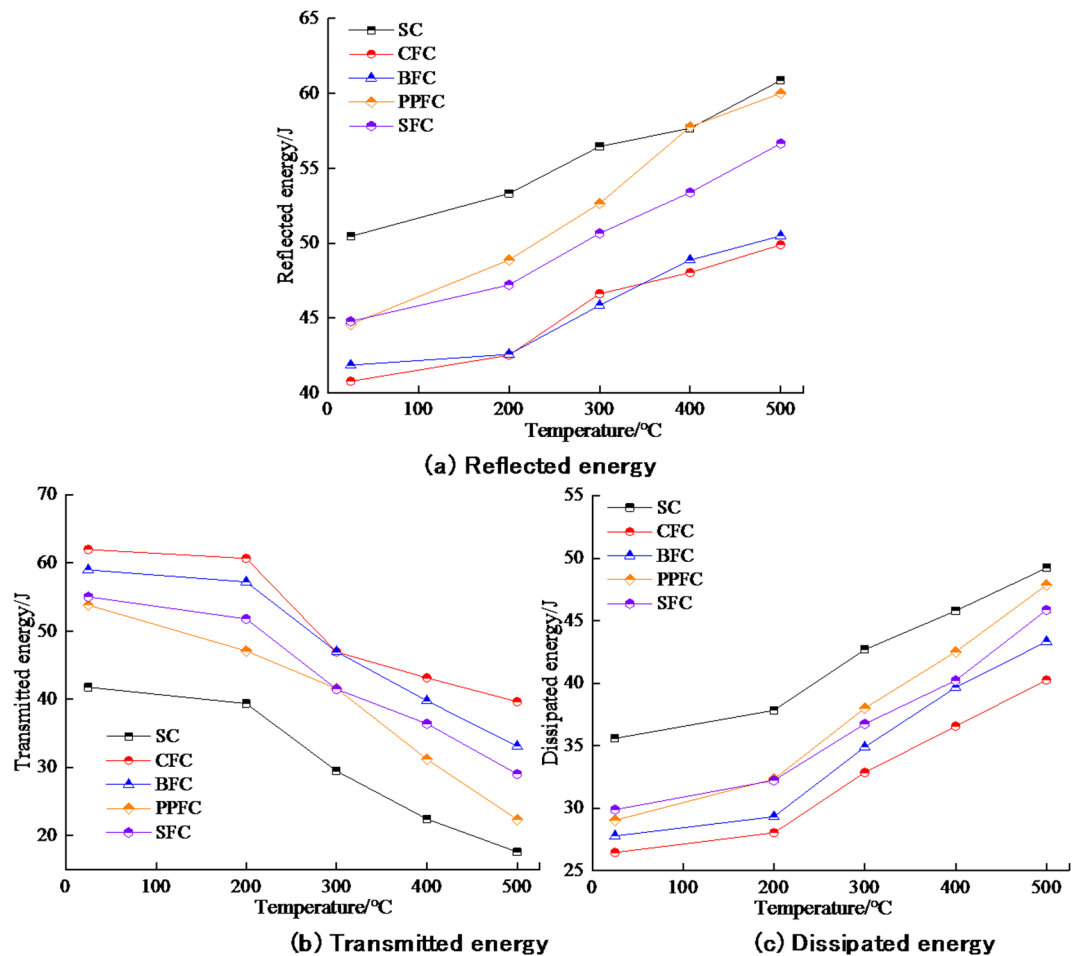


Fig. 10. The relationship between energy and temperature.

its load-bearing capacity. On the other hand, the fiber itself has high tensile strength. When the specimen is subjected to external force, the fiber is also subjected to load, and its high tensile strength can effectively absorb and consume external energy when subjected to external load. At the same time, the good compatibility between fiber and cement-based material makes it can be closely bonded, thus increasing the deformation and failure resistance of the specimen itself. Figure 13 shows that with the increase of temperature, the degree of breakage of SC and CFC increases continuously, the diameter of the broken block of the specimen decreases continuously, the number of broken blocks increases continuously, and the higher the temperature, the greater the degree of breakage. As a result of high temperature, the internal water of the test specimen is continuously lost, and the bonding between aggregates decreases. Under external impact load, the test specimen is more prone to damage, and the energy dissipation density of the test specimen is also increasing. The degree of breakage of the specimen was quantitatively characterized based on three-dimensional fractal characteristics⁴⁴, as shown in Fig. 14. The fracture fractal dimension of SC under impact load is always higher than that of fiber concrete. Compared with the fracture fractal dimension of SC specimen at 25°C, the fracture fractal dimension of CFC, BFC, PPFC and SFC specimen decreases by 8.98%, 7.76%, 3.67% and 1.63%, respectively. With the increase of temperature, the fractal dimension of the specimen is increasing. Among them, at 0°C, 200°C, 300°C, 400°C and 500°C, the fracture fractal dimension of SC specimen is 2.45, 2.49, 2.57, 2.71 and 2.85, respectively, and the increase of PPFC fractal dimension is the most significant, which is consistent with the growth trend of crushing energy density.

Therefore, the effect of high temperature will cause damage to the performance of concrete materials and buildings. The microscopic form is the increase of porosity and the increase of fractal dimension of pore structure, and the macroscopic form is the decrease of strength and the increase of fracture degree. The addition of fiber can not only enhance the internal structure and impact resistance of concrete material, but also enhance its high temperature resistance. CF has the most significant effect on the high temperature resistance of concrete materials, followed by BF, SF and PPF. This study can provide test parameters for the safe use of concrete buildings in high temperature environment.

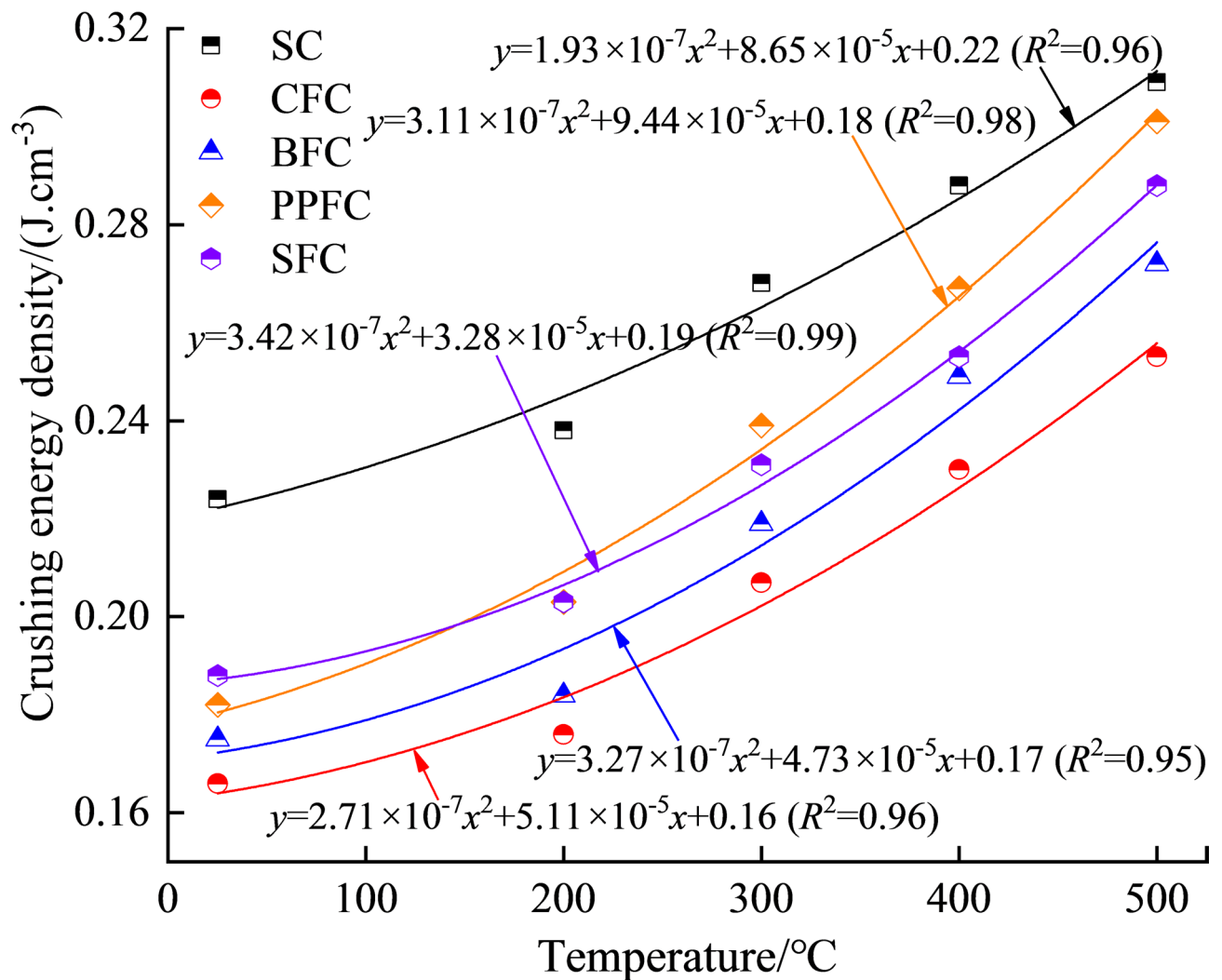


Fig. 11. Change law of crushing dissipation energy.

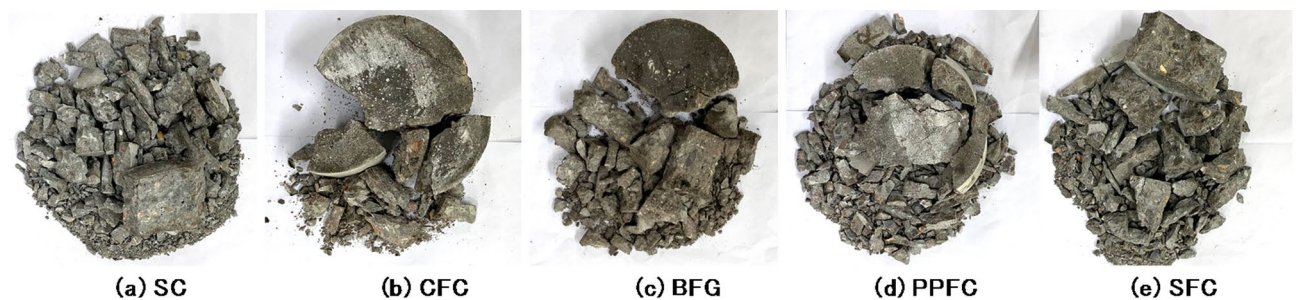


Fig. 12. Change law of crushing dissipation energy.

Conclusion

- (1) Compared with the main peak value of plain concrete at 25°C, the main peak value of CFC, BFC, PPFC and SFC specimens decreased by 38.11%, 35.85%, 26.39% and 23.00%, respectively. The addition of fiber could reduce the porosity of specimens, while the effect of high temperature would increase the porosity of specimens.
- (2) Compared with the porosity of the specimens at 25°C, the increase in porosity of plain concrete at 200°C, 300°C, 400°C and 500°C was 14.53%, 26.59%, 45.36% and 81.23% respectively, and the increase in porosity of carbon fiber concrete was 35.46%, 52.48%, 94.33% and 141.84% respectively. With the increase of tem-

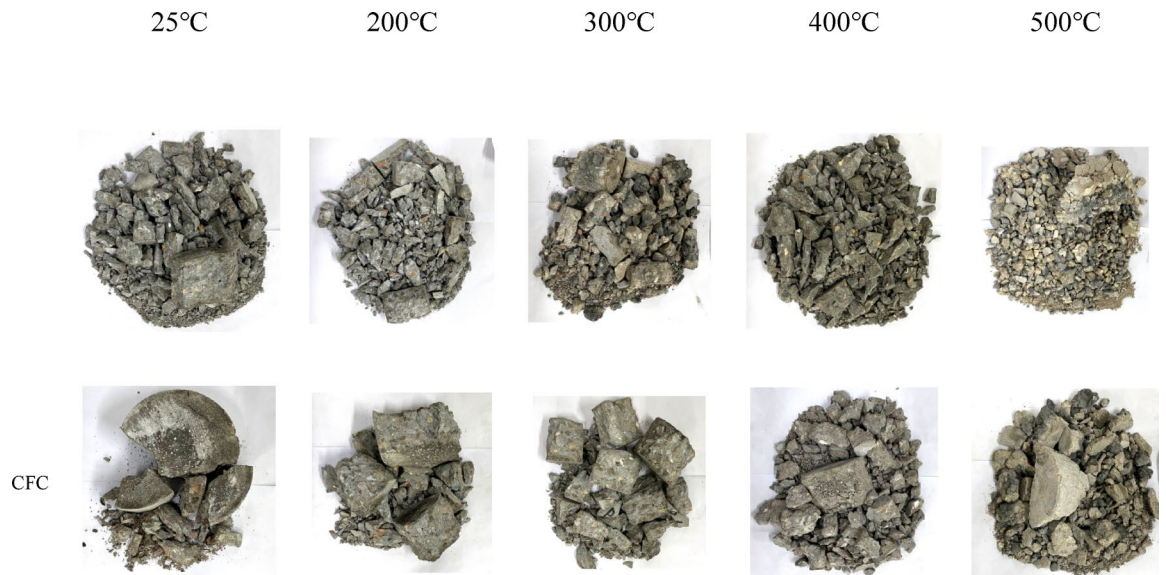


Fig. 13. Influence of high temperature on fracture morphology of specimens.

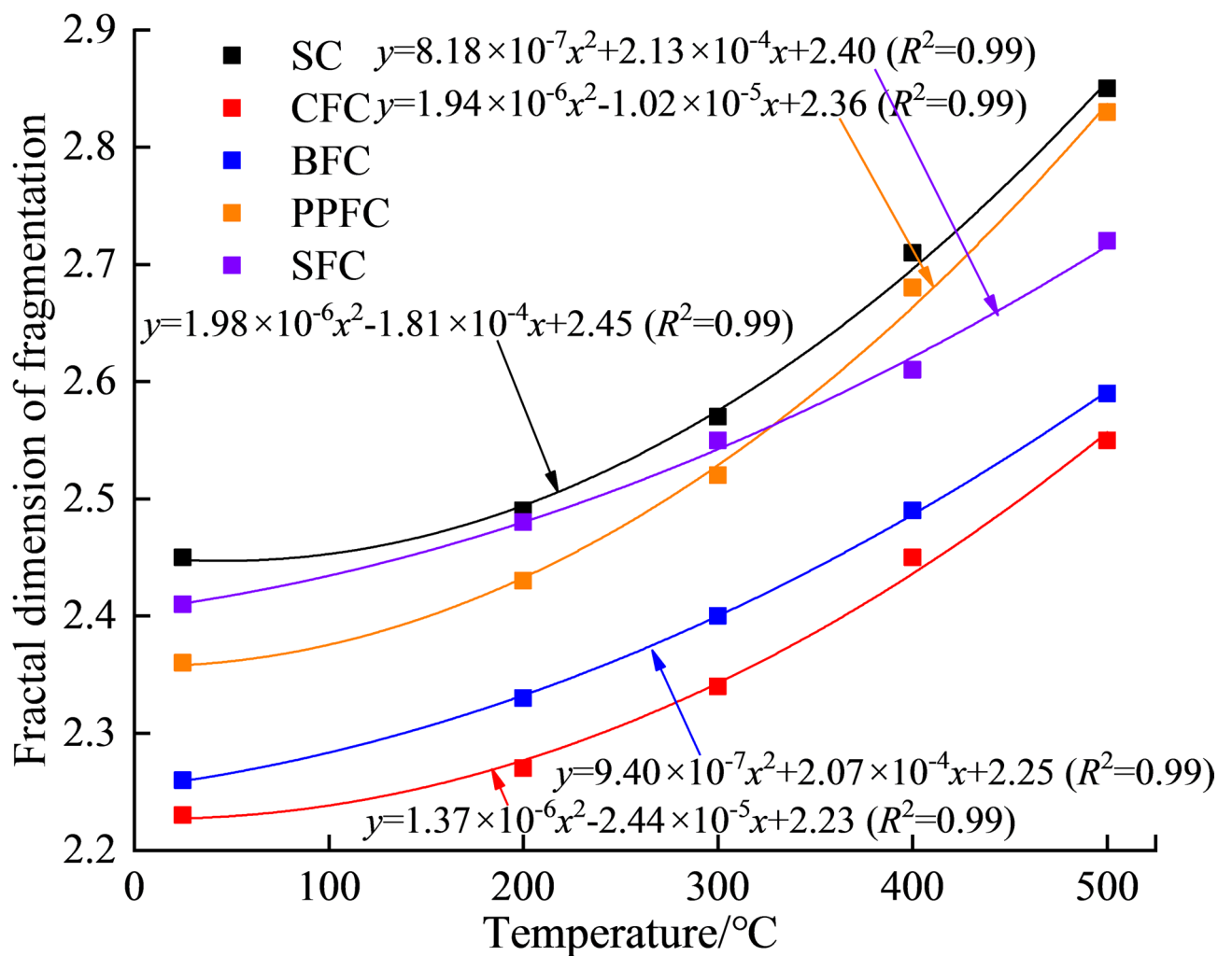


Fig. 14. The relationship between the fractal dimension of fragmentation and temperature.

perature, $D_{\min}$ and $D_{\max}$ of five kinds of concrete specimens increase continuously, and $D_{\max}$ of specimens is always higher than $D_{\min}$. The complexity of the large pore structure inside the specimen is higher. The increase in temperature leads to an increase in the internal porosity and pore diameter of the specimens, as well as an increase in the complexity of the structure.

- (3) At 25°C, the peak stress of CFC, BFC, PPFC and SFC specimens is 1.44, 1.37, 1.13 and 1.17 times that of plain concrete. The addition of fiber effectively improves the compressive strength of concrete specimens, and CF has the highest effect on the strength of concrete. Compared with the peak stress of the specimens at 25°C, the decreases in the peak stress of the SC specimens at 200°C, 300°C, 400°C and 500°C were 5.03%, 12.40%, 25.30% and 44.04% respectively, and the decreases in toughness were 5.56%, 16.67%, 29.63% and 44.44% respectively. The peak stress and toughness of PPFC specimens decreased continuously under high temperature, and the degradation degree of PPFC specimens was the highest.
- (4) The fractal dimensions of the SC specimens at 0°C, 200°C, 300°C, 400°C and 500°C were 2.45, 2.49, 2.57, 2.71 and 2.85 respectively. High temperature will cause damage to the performance of concrete materials and buildings. The microcosmic manifestation is the increase of porosity and fractal dimension of pore structure, while the macroscopic manifestation is the decrease of strength and the increase of fracture degree. The addition of fiber can enhance the deformation resistance of concrete material, and CF can improve the performance of concrete material more significantly.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

S.D. and H.C. wrote the main manuscript text and E.F., Z.C. and Z.F. prepared figures. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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